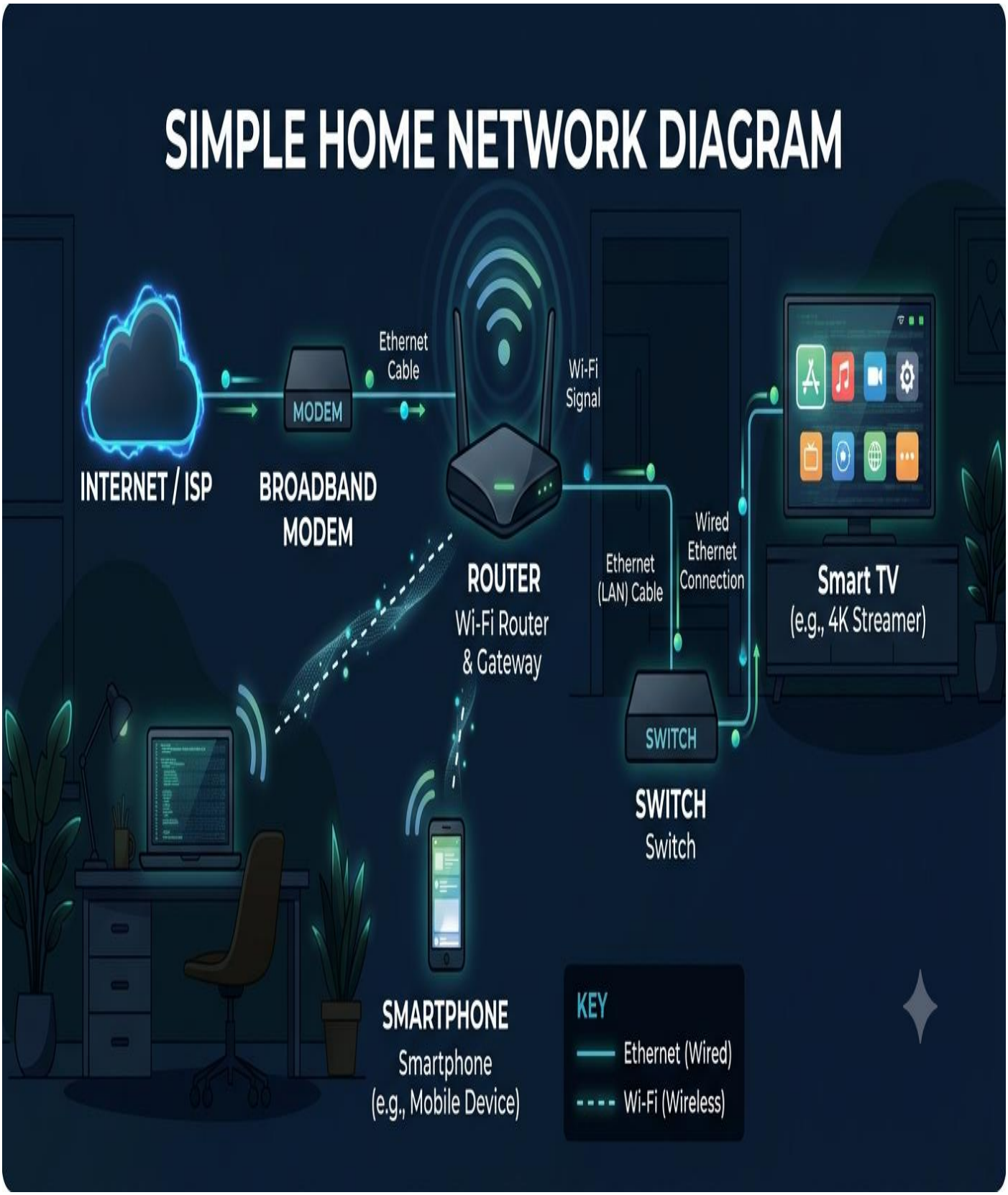
8/10/2026

Assignment-6

Created By
~pintu mer

1. **Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.**



2. **Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).**

❖ **Network Devices and Real-World Uses**

Network Device	Main Function	Real-World Use Case
Router	Connects different networks and provides internet access.	We use a Wi-Fi router to access YouTube, Instagram, and WhatsApp on our mobile phones.
Switch	Connects multiple devices within a LAN.	In a gaming zone, a switch connects multiple PCs and gaming consoles for LAN gaming.
Hub	Connects multiple devices in a LAN and sends data to all connected ports.	In an old computer lab, a hub can connect multiple computers to share network resources.
Access Point (AP)	Provides Wi-Fi connectivity to wireless devices.	In a college, an access point provides Wi-Fi for students to browse websites and watch YouTube videos.

3. **Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).**

Scenario: Ordering Food on Zomato

- **When I order food using the Zomato app on my smartphone, data travels through the following network devices.**

1. **Access Point (AP): My phone connects to Wi-Fi through the access point wirelessly.**
2. **Router: The router sends my order data from my home network to the internet.**
3. **Switch: Switches help transfer data between devices and servers within the ISP or data center network.**

4. **Internet: My order request reaches Zomato's server, which processes the order and sends it to the restaurant.**
5. **Response: The order confirmation travels back through the internet and router to my phone.**

Example: When I click Place Order on Zomato, my order reaches the Zomato server, and I receive an order confirmation on my phone.

4. **Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.Hint: Think about devices like switches, powerline adapters, or WiFi extenders.**

Answer: Two Ways to Connect a Gaming PC to Wi-Fi Router

1. Powerline Adapter

- **It uses the existing electrical wiring in your home to carry network data.**
- **Connect one adapter to the router using an Ethernet cable and plug it into a power socket.**
- **Plug the second adapter near your gaming PC and connect it using an Ethernet cable.**
- **Example: Play online games like PUBG with a wired network connection without running a long Ethernet cable.**

2. Wi-Fi Extender with an Ethernet Port

- **It receives the Wi-Fi signal from your router and extends the network coverage.**
- **Place it near your gaming PC, where the Wi-Fi signal is strong enough.**
- **Connect the extender's Ethernet port to your gaming PC.**
- **Example: Access the internet on your gaming PC without installing a Wi-Fi card.**

Simple diagram

Option 1: Powerline Adapter



Router → Powerline adapters → Ethernet cable → Gaming PC

Option 2: Wi-Fi Extender



Router → Wi-Fi signal → Extender → Ethernet cable → Gaming PC